

Number theory

a. Divisibility

Definitions and properties

- **Def.** We say “a divides b”, or $a|b \Leftrightarrow \exists d, a \cdot d = b$
 - Also “a is a divisor of b” or “b is a multiple of a”
 - Notice this means, by def., we have $\forall n, n|0$
 - Because $\forall n \ n \cdot 0 = 0$
- **Thm:** $\forall a, b_1, b_2, a|b_1 \wedge a|b_2 \Rightarrow a|(b_1 + b_2)$
- **Pf:** By definition we know
 - There exists d_1, d_2 , so that $a \cdot d_1 = b_1$ and $a \cdot d_2 = b_2$
 - So, plugging in: $b_1 + b_2 = a \cdot d_1 + a \cdot d_2$
 - And this implies that $b_1 + b_2 = a(d_1 + d_2)$
 - QED (by definition of divides)

Some easy questions about divisibility

- **Question 1:** Does $d|a$ and $d|b \Rightarrow d|(a + b)$?
- **Question 2:** Does $d|(a+b) \Rightarrow d|a$ and $d|b$?
 - Ex. $3|(10+11)$ but 3 doesn't divide either 10 or 11

Properties of “divides”

- **Def.** We say a binary relation R is transitive iff:
 - $\forall a, b, c, R(a, b) \wedge R(b, c) \Rightarrow R(a, c)$
- **Ex:** $=, <, \leq$ (why?)
- **Thm.** Divisibility is transitive
 - i.e., $\forall a, b, c, a|b \wedge b|c \Rightarrow a|c$
- **Pf.** By def. $\exists d_1, a \cdot d_1 = b$ and $\exists d_2, b \cdot d_2 = c$
 - But then by plugging in for b , $a \cdot d_1 \cdot d_2 = c$
 - QED (via $d_1 \cdot d_2$).

Properties of “set of divisors”

- **Def.** The set of divisors of n is denoted $Div(n) = \{a, a|n\}$
- **Ex:** $Div(6) = \{\pm 1, \pm 2, \pm 3, \pm 6\}$
- **Thm.** $a|b \Rightarrow Div(a) \subseteq Div(b)$
- **Pf.** Suffices to show $\forall a' \in Div(a), a'|b$
 - Since $a|b$ we know $\exists d_1, a \cdot d_1 = b$
 - Now since $a' \in Div(a) \Rightarrow \exists d_2, a' \cdot d_2 = a$
 - Plugging in for a , $a' \cdot d_2 \cdot d_1 = b$
 - QED (via $d_2 \cdot d_1$)

Greatest Common Divisor and Division Thm.

- **Def.** If m and n are nonzero integers, then $\text{GCD}(m,n)$ is the largest element of $\text{Div}(m) \cap \text{Div}(n)$
- **Ex.** $\text{GCD}(5,15)=5$
- **Thm.** (The Division Thm.) $\forall n, d > 0, \exists! q, r$ so that $n = d \cdot q + r$ and $0 \leq r < d$
 - Note: we call q the quotient, d the divisor, and r the remainder
- **Pf. 1** (Existence, nonconstructive) $n - d \cdot n = n(1-d)$
 - Consider the set $A = \{n - d \cdot q \mid q \in \mathbb{Z}\}$ “The remainder set”
 - Since $d \neq 0$, q can be positive or negative, A must have a non-negative member
 - Therefore, A must have a *least nonnegative* member, called r
 - Now $r < d$ (why?)
 - because otherwise by subtraction, $0 \leq r - d < r$ would be a non-negative element of A , but this contradicts the minimality of r
 - So Let $q \in \mathbb{Z}$ witness that $r \in A$, then notice $r = n - d \cdot q \Rightarrow n = d \cdot q + r$